

1. Administrative & Study Identification

Full Study Title: Phase III Study of Rilvegostomig in Combination With Bevacizumab With or Without Tremelimumab as First-line Treatment of Hepatocellular Carcinoma **Official Title:** A Phase III, Randomised, Open-label, Sponsor-blinded, Multicentre Study of Rilvegostomig in Combination With Bevacizumab With or Without Tremelimumab as First-line Treatment in Patients With Advanced Hepatocellular Carcinoma **Short Title/Acronym:** ARTEMIDE-HCC01 **Protocol Number:** D7029C00001 **NCT Number:** NCT06921785 **Sponsor Name:** AstraZeneca **Investigational Product:** Rilvegostomig (AZD2936), Bevacizumab (Trade names: Avastin, Vegzelma, Alymsys, Zirabev, Mvasi), Tremelimumab **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To demonstrate the efficacy of rilvegostomig + bevacizumab + tremelimumab (Arm A) relative to atezolizumab + bevacizumab (Arm C) by assessing Overall Survival (OS) in participants with advanced Hepatocellular Carcinoma (HCC). **Secondary Objectives:** To evaluate OS (Arm B vs Arm C), Objective Response Rate (ORR), Duration of Response (DoR), Progression-Free Survival (PFS), Safety/tolerability (occurrence of adverse events), and overall quality of life.

2.2 Background & Rationale HCC is the most common liver cancer globally. While first-line treatment comprises immunotherapy combinations, outcomes often remain poor due to diverse tumor immune-evasive mechanisms. Rilvegostomig is a novel bispecific anti-PD-1/anti-TIGIT antibody designed to enhance antitumor activity by overcoming immune-inhibitory signaling. This study assesses whether combining dual or triple immune checkpoint blockade (PD-1, TIGIT, CTLA-4) with anti-VEGF targeted therapy can meaningfully improve outcomes versus the current standard of care.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Trial Status:** RECRUITING **Control Type:** Active Comparator (Atezolizumab + Bevacizumab) **Blinding:** Open-label, Sponsor-blinded **Randomization:** 1:1:1 (Stratified by macrovascular invasion/extrahepatic spread, liver disease etiology, alpha-fetoprotein level, and PD-L1 expression)

3.2 Planned Enrollment & Duration Target \$\$\$: 1,220 participants
Number of Sites: Multicenter, Global **Estimated Study Start:** 06-May-2025 **Estimated Primary Completion:** 16-Mar-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Locally advanced or metastatic and/or unresectable HCC. 2. WHO/ECOG performance status of 0 or 1. 3. BCLC stage B (that is not eligible for locoregional therapy) or stage C, with Child-Pugh Score class A. 4. At least one measurable target lesion. 5. Adequate organ and bone marrow function measured during the screening period. 6. Must not have received prior systemic therapy for intermediate, advanced, or metastatic HCC. 7. Disease that is not amenable to curative surgical and/or locoregional therapies. (For participants who received locoregional therapy for HCC, locoregional therapy must have been completed $\geq$ 28 days prior to the baseline scan).

4.2 Exclusion Criteria 1. Co-infected with HBV and HCV. 2. Known fibrolamellar HCC, sarcomatoid HCC, or mixed cholangiocarcinoma and HCC. 3. Prior treatment with anti-CTLA-4 and/or anti-TIGIT. 4. Any evidence of uncontrolled intercurrent diseases, active or prior documented autoimmune or inflammatory disorders requiring chronic treatment with steroids or other immunosuppressive treatment. 5. History of another primary malignancy. 6. Persistent toxicities caused by previous anti-cancer therapy excluding alopecia, not yet improved to Grade $\leq$ 1 or baseline. 7. Clinically meaningful ascites, pleural effusion, or pericardial effusion requiring non-pharmacologic intervention to maintain symptomatic control within 6 months prior to the first scheduled dose. 8. History of active primary immunodeficiency or active infection, or history of hepatic encephalopathy. 9. Current or recent (within 10 days of first dose) use of aspirin ($\geq$ 325 mg/day), treatment with dipyridamole, ticlopidine, clopidogrel, cilostazol, or use of full-dose oral or parenteral anticoagulants or thrombolytic agents for therapeutic purposes. 10. Central nervous system metastases or spinal cord compression. 11. Radiotherapy within 28 days and abdominal/pelvic radiotherapy within 60 days prior to initiation of study treatment, except palliative radiotherapy to bone lesions within 7 days.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm A:** Rilvegostomig + Bevacizumab + Tremelimumab * **Arm B:** Rilvegostomig + Bevacizumab * **Arm C:** Atezolizumab + Bevacizumab * **Safety Lead-in Phase:** Single-arm period applied prior to randomisation (N=20) evaluating Rilvegostomig + Bevacizumab + Tremelimumab **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Until disease progression or unacceptable toxicity (with overall survival tracked up to approximately 6 years).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive therapies and symptom management interventions. **Prohibited:** Prior systemic therapies for intermediate, advanced, or metastatic HCC, and concurrent investigational agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Scan (locoregional therapies must be \$ \ge\$ 28 days prior)
Baseline	Day 0	Randomization, First Dose Administration
Interim	Regular Cycles	Safety Labs, Efficacy & Tumor Assessments, Infusions
End of Study	Up to 6 Years	Disease progression evaluation, Exit Interview, Long-term Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall Survival (OS) - defined as the time from randomization until the date of death due to any cause (specifically comparing Arm A to Arm C). **Measurement Method:** Survival tracking up to approximately 6 years.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) (Arm B vs Arm C) * Objective Response Rate (ORR), Duration of Response (DoR), and Progression-Free Survival (PFS)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Occurrence of AEs and SAEs evaluated and graded according to the revised NCI CTCAE v5.0 up to approximately 6 years. **Serious Adverse Events (SAEs):** Routine SAE monitoring protocol typical of Phase III oncology trials. **Data Monitoring Committee (DMC):** A 20-patient, single-arm safety lead-in period is leveraged to rigorously evaluate the safety and tolerability of combining rilvegostomig, bevacizumab, and tremelimumab prior to initiating the randomized phase.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy assessments (OS, PFS). **Sample Size Justification:** Target enrollment of \$1,220\$ is powered to detect statistically significant improvements in OS between the experimental arms (A and B) versus the active control arm (C). **Handling of Missing Data:** Standard oncology data censoring practices for subjects lost to survival follow-up or PFS evaluation.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard global clinical oncology monitoring to oversee site activity, AE tracking, and drug accountability.

Audit Trail: Centralized eCRF data capture and documentation tracking mechanisms.

11. Study Identifiers & Governance

Primary ID: D7029C00001 **NCT Number:** NCT06921785 **Posting ID:** 277162575593087325 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ARTEMIDE-HCC01 **Official Regulatory Title:** A Phase III, Randomised, Open-label, Sponsor-blinded, Multicentre Study of Rilvegostomig in Combination With Bevacizumab With or Without Tremelimumab as First-line Treatment in Patients With Advanced Hepatocellular Carcinoma

12. Clinical Framework (Hepatocellular Carcinoma Specific)

Primary Condition: Hepatocellular Carcinoma (HCC) **Diagnosis Threshold:** Locally advanced or metastatic and/or unresectable HCC (BCLC stage B or C); Child-Pugh Score class A **Medical Methodology:** First-line systemic immune checkpoint blockade combined with anti-angiogenic therapy **Optimization Target:** Maximum antitumor activity, delayed disease progression, and prolonged Overall Survival (OS)

13. Study Procedures & Methodology

Management Pathway: First-line systemic clinical pathway for advanced HCC **Education Protocol (HCP):** Site initiation and training on administration of investigational bispecific antibodies and immune-mediated adverse event management **Education Protocol (Patient):** Trial onboarding, understanding of infusion schedules, and toxicity reporting **Quality Audit Mechanism:** Standardized radiologic tumor imaging review and independent safety tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 6 years of tracking **Onsite Visit Cadence:** Regular, aligned with IV treatment cycles and required tumor imaging intervals **Remote Monitoring Frequency:** Intermittent routine survival follow-up post-progression

Checklist Review Interval: Toxicity/AE assessment prior to the administration of each treatment cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				